Baleashvar Kasiviswanathan

PH:+91 7904329613 | LinkedIn | GitHub | Portfolio | baleashvar@gmail.com

Professional Summary

Cloud DevOps, Reliability, and Software Engineer with 3+ years of experience engineering scalable cloud infrastructure, real-time streaming architectures, and developer tooling across aviation domain. Proven track record in slashing data latency by 99%, optimizing cloud compute costs, and automating mission-critical workflows using AWS, Python, and Kafka. Deep practical expertise in LLM agent orchestration, prompt engineering, MCP. and semantic caching layers.

Core Competencies

- Cloud & Infrastructure: AWS (Lambda, ECS, EKS, DynamoDB, S3, IAM, CloudWatch), Docker, Kubernetes, CI/CD, Linux / Bash
- Streaming & Data Engineering: Apache Kafka, DynamoDB, PostgreSQL, MySQL
- Programming & Frameworks: Python (FastAPI, Boto3), Node.js, Express.js, TypeScript/JavaScript, React.js
- AI & Machine Learning: LLM Agents, Prompt Engineering, Semantic Caching (sentence-transformers), Generative Adversarial Networks (SRGAN), TensorFlow
- Reliability & Operations: Incident Response, Root Cause Analysis (RCA), Performance Tuning, Distributed Tracing, API Testing (Postman, pytest)

Professional Experience

Tata Consultancy Services (TCS) | Chennai, India

Cloud DevOps & Reliability Engineer (British Airways Account) **AI-Driven Operations | *Sep 2023 – Present***

- **Mission-Critical Incident Management:** Owned L3 operational support, ticket lifecycle management, and incident triage for core aviation services, strictly maintaining >99.95% availability across demanding enterprise SLAs.
- **Incident Bridge Leadership & MTTR Reduction:** Served as technical incident bridge responder during high-severity production P1/P2 outages; leveraged Datadog APM distributed traces and CloudWatch logs to rapidly locate system bottlenecks and reduce MTTR.
- **Proactive APM & Anomaly Detection:** Configured Datadog machine learning anomaly detection and metric threshold monitors across microservices and message queues, identifying latent resource contention and memory leaks prior to end-user degradation.
- **Cloud Security & Vulnerability Remediation:** Handled ongoing security patch management and vulnerability remediation across cloud compute instances and container environments; hardened IAM roles and policies under formal Change Management controls.
- **Runbook & Operational Automation:** Engineered custom Python (Boto3) and Bash utilities to automate manual operational procedures, diagnostic log dumps, and repetitive health checks, cutting manual intervention overhead.
- **Infrastructure as Code & CI/CD Delivery:** Implemented and rolled out cloud configuration updates and vulnerability fixes across AWS environments using modular Terraform code and GitHub Actions pipelines in compliance with Change Advisory Board (CAB) protocols.
- **ITIL Problem Management & RCA:** Spearheaded post-incident reviews (RCA), operational risk assessments.

Technical Projects

Semantic Cache for LLMs (semantic-cache-llm) | *Python, sentence-transformers, FAISS*

- Developed an open-source semantic caching layer for Large Language Models to detect redundant queries and eliminate unnecessary API round-trips.
- Implemented local embedding generation (all-MiniLM-L6-v2) and cosine similarity thresholds, reducing API inference latency and token consumption by up to 40%.

JSONLab (jsonlab.xyz) | *React, TypeScript, Cloudflare/Vercel*

- Designed and deployed a browser-based developer utility platform for rapid formatting, schema validation, and conversion of high-volume JSON payloads.

Education

- Bachelor of Engineering (B.E.) in Electronics and Communication Engineering
  Meenakshi Sundararajan Engineering College, Anna University, Chennai
  CGPA: 8.28 / 10.0 | Graduated: 2022

Certifications & Research

- AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner
- **Research Publication**:*"Planetary Image Improvisation Using Super-Resolution Generative Adversarial Networks (SRGAN)"* (TensorFlow-based image enhancement for deep-space astronomy imagery).